# BLINKIT GROCERY OUTLET PERFORMANCE ANALYSIS

A Data Analytics Project Using Microsoft Power BI

| | |
|---|---|
| **Dataset** | **BlinkIT Grocery Data.csv** |
| **Records and fields** | 8523 records • 12 columns |
| **Analysis period** | Not Specified |
| **Tools** | Microsoft Power BI, CSV dataset, data profiling |

| Column1 | Item Fat Content | Item Identifier | Item Type | Outlet Establishment Year | Outlet Identifier | Outlet Location Type | Outlet Size | Outlet Type | Item Visibility | Sales | Rating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 685 | Low Fat | NCP18 | Household | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.028760013 | $151.4708 | 4 |
| 806 | Low Fat | FDR48 | Baking Goods | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.132248069 | $150.5024 | 4 |
| 807 | Low Fat | FDO12 | Baking Goods | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.055241242 | $196.5452 | 4 |
| 808 | Low Fat | FDS47 | Breads | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.12961476 | $86.1856 | 4 |
| 809 | Low Fat | FDZ27 | Dairy | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.017253305 | $51.435 | 4 |
| 810 | Low Fat | DRE03 | Dairy | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.024363939 | $46.3718 | 4 |
| 811 | Low Fat | FDB08 | Fruits and Vegetables | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.03127929 | $160.3578 | 4 |
| 812 | Low Fat | FDB44 | Fruits and Vegetables | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0 | $212.4586 | 4 |
| 813 | Low Fat | FDE57 | Fruits and Vegetables | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.036490369 | $140.8154 | 4 |
| 814 | Low Fat | FDN20 | Fruits and Vegetables | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.026329989 | $169.2474 | 4 |
| 815 | Low Fat | NCD19 | Household | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.013253936 | $56.4614 | 4 |
| 959 | Low Fat | NCV17 | Health and Hygiene | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.01619866 | $129.2626 | 4 |
| 1088 | Low Fat | FDK48 | Baking Goods | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.037845086 | $75.5354 | 4 |
| 1089 | Low Fat | FDH48 | Baking Goods | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.06082246 | $85.154 | 4 |
| 1090 | Low Fat | FDZ38 | Dairy | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.008046275 | $173.7422 | 4 |
| 1091 | Low Fat | DRH39 | Dairy | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.093214498 | $74.667 | 4 |
| 1092 | Low Fat | FDK04 | Frozen Foods | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.052607932 | $56.3588 | 4 |
| 1093 | Low Fat | FDQ16 | Frozen Foods | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.041974489 | $108.6912 | 4 |
| 1094 | Low Fat | FDX20 | Fruits and Vegetables | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.04280099 | $228.372 | 4 |
| 1095 | Low Fat | FDC44 | Fruits and Vegetables | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.173573164 | $114.4518 | 4 |
| 1096 | Low Fat | DRK23 | Hard Drinks | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.072383777 | $251.504 | 4 |
| 1097 | Low Fat | DRO47 | Hard Drinks | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.112859454 | $114.486 | 4 |
| 1098 | Low Fat | DRJ23 | Hard Drinks | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.041904578 | $188.1872 | 4 |
| 1099 | Low Fat | NCL53 | Health and Hygiene | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.036439878 | $175.3028 | 4 |
| 1100 | Low Fat | NCI17 | Health and Hygiene | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.1442339 | $95.441 | 4 |
| 1101 | Low Fat | NCL29 | Health and Hygiene | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.114583922 | $156.4604 | 4 |
| 1102 | Low Fat | NCA53 | Health and Hygiene | 2020 | OUT017 | Tier 2 | Small | Supermarket Type1 | 0.009934335 | $47.2034 | 4 |

Table: blinkit_cleaned_dataset (8,523 rows) Column: Rating (5 distinct values)

# 1. Executive Summary

This project analyzes Blinkit's sales dataset containing **8,523 item records** to evaluate sales performance across product categories, fat content, outlet establishment years, outlet sizes, location tiers, and outlet types. The objective is to identify key sales drivers, understand differences in outlet performance, and develop data-driven business recommendations. The dataset recorded approximately **1.20 million in total sales**, with average sales of **140.99 per item record** and an average customer rating of **3.97 out of 5**. The analysis indicates that Low Fat products contribute the largest share of sales, while Fruits and Vegetables and Snack Foods are the leading item categories by total sales. Medium-sized outlets and Tier 3 locations contribute the highest sales among their respective segments. Supermarket Type1 outlets dominate overall sales, accounting for approximately 65.54% of the total. These findings suggest opportunities to strengthen inventory planning for high-performing categories, optimize product availability across outlet segments, investigate lower-performing categories, and identify practices associated with stronger customer ratings. Overall, this analysis provides a foundation for improving sales management and outlet-level decision-making. Further evaluation of profitability, operating costs, inventory availability, and sales performance per outlet would help determine which opportunities can deliver sustainable business growth.

# 2. Business Objectives

1. Analyse **Sales Performance:** Evaluate overall sales and identify the key contributors to total sales across different product categories and outlet segments.
2. Evaluate **Product Categories:** Identify high-performing and low-performing item types to understand product demand and opportunities for assortment optimization.
3. Analyse **Fat Content Preferences:** Compare sales of Low Fat and Regular products to understand their contribution to overall sales and related customer ratings.
4. Assess **Outlet Performance:** Compare sales across different outlet types and establishment years to identify variations in performance.
5. Evaluate **Outlet Size:** Examine sales distribution across small, medium, and high-sized outlets to understand the relationship between outlet size and sales contribution.
6. Analyse **Location-Based Sales:** Compare sales across Tier 1, Tier 2, and Tier 3 locations to identify geographic sales patterns and potential growth opportunities.
7. Monitor **Key Performance Indicators (KPIs):** Evaluate total sales, average sales, item record counts, and average customer ratings across outlet types to support performance monitoring.
8. Develop **Business Recommendations:** Translate analytical findings into practical recommendations for inventory planning, product assortment, outlet performance improvement, and customer satisfaction.

# 3. Dataset Overview

The dataset used for this project, **blinkit_cleaned_dataset.csv**, contains 8,523 records and 12 columns related to product sales, item characteristics, customer ratings, and outlet attributes. It provides a foundation for analyzing sales performance across different product categories and retail outlet segments.

| Area | Fields / examples | Purpose |
|---|---|---|
| **Product Information** | Item Identifier, Item Type, Item Fat Content | Analyze sales by product category and fat content. |
| **Sales Metrics** | Sales | Measure total sales and average sales per record. |
| **Customer Feedback** | Rating | Evaluate average customer ratings. |
| **Outlet Information** | Outlet Identifier, Outlet Type, Outlet Size | Compare sales across outlet formats and sizes. |
| **Location Information** | Outlet Location Type | Analyze sales across Tier 1, Tier 2, and Tier 3 locations. |
| **Establishment Details** | Outlet Establishment Year | Examine sales distribution by establishment year. |
| **Product Availability** | Item Visibility | Explore the relationship between product visibility and sales. |

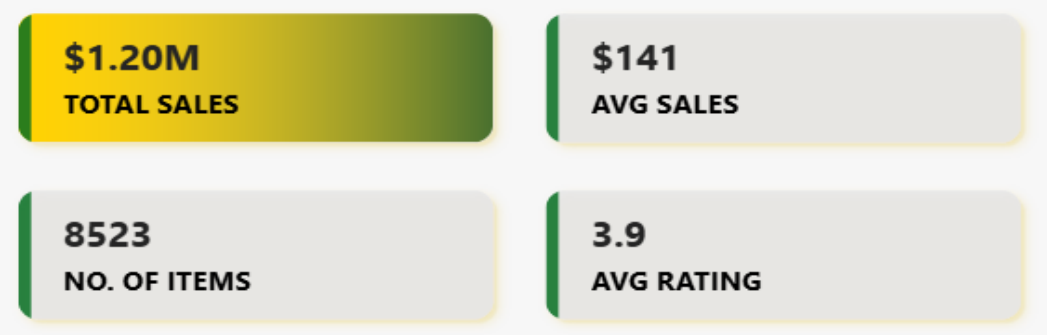
# 4. Data Quality and Preparation

Firstly, the data gone through simple checks, which can be seen directly from human eyes like whether the file was excel file or the csv file. Then we move to jupyter notebook to load the dataset using the Python's Pandas library, then we checked the column names, number of columns. Then we checked unique values of all required columns. Item_weight column is irrelevant for this analysis, so dropped the irrelevant columns. Then we found the column Item Fat Content have messy unique Values. Actually, it must have only two values, but having four values. So, replacing unwanted unique values to normal form we got the final cleaned dataset ready for visualization. We load it to Power BI and start visualizations.

# 5. KPI Definitions and Results

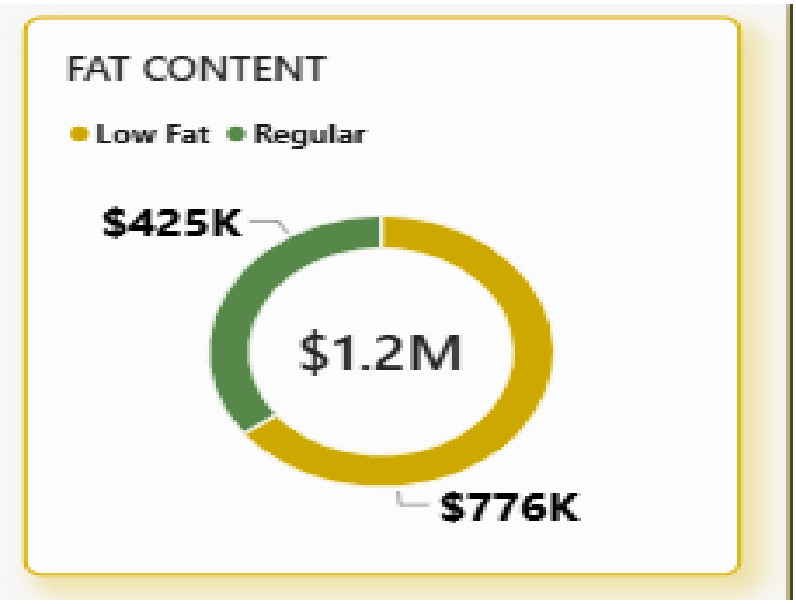
| KPI | Definition | Dataset result |
|---|---|---|
| **Total Sales** | Sum Of sales | $1.20 M |
| **Average sales** | Average of all sales | $141 |
| **No. of Item** | Count of the items | 8523 |
| **Average Rating** | Average of non-missing Customer Rating | 3.9 / 5 |

For context, all the 8523 records and their Total sales $ 1.20M, Average sales of all the recorded sales are $141, for the Number of items i.e. the count of the total items is 8523 same as the total number of records. Finally, the average rating of the customer ratings provided is recorded to be 3.9 out of the 5.



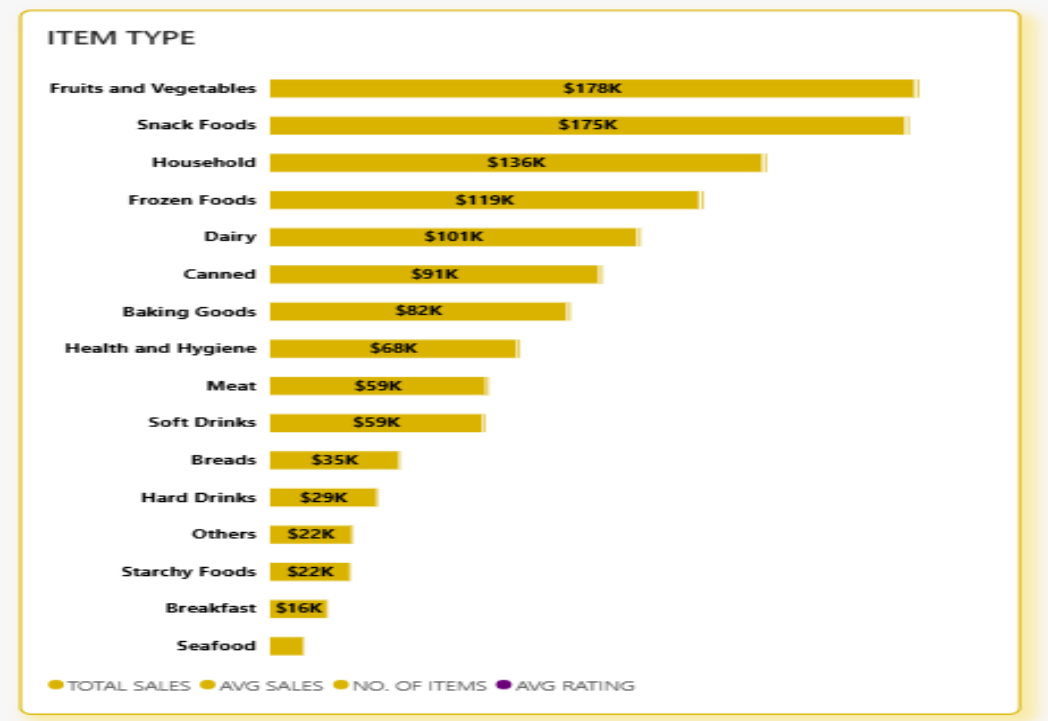

# 6. Dashboard Visualizations and Findings

## 6.1 Total sales by FAT Content



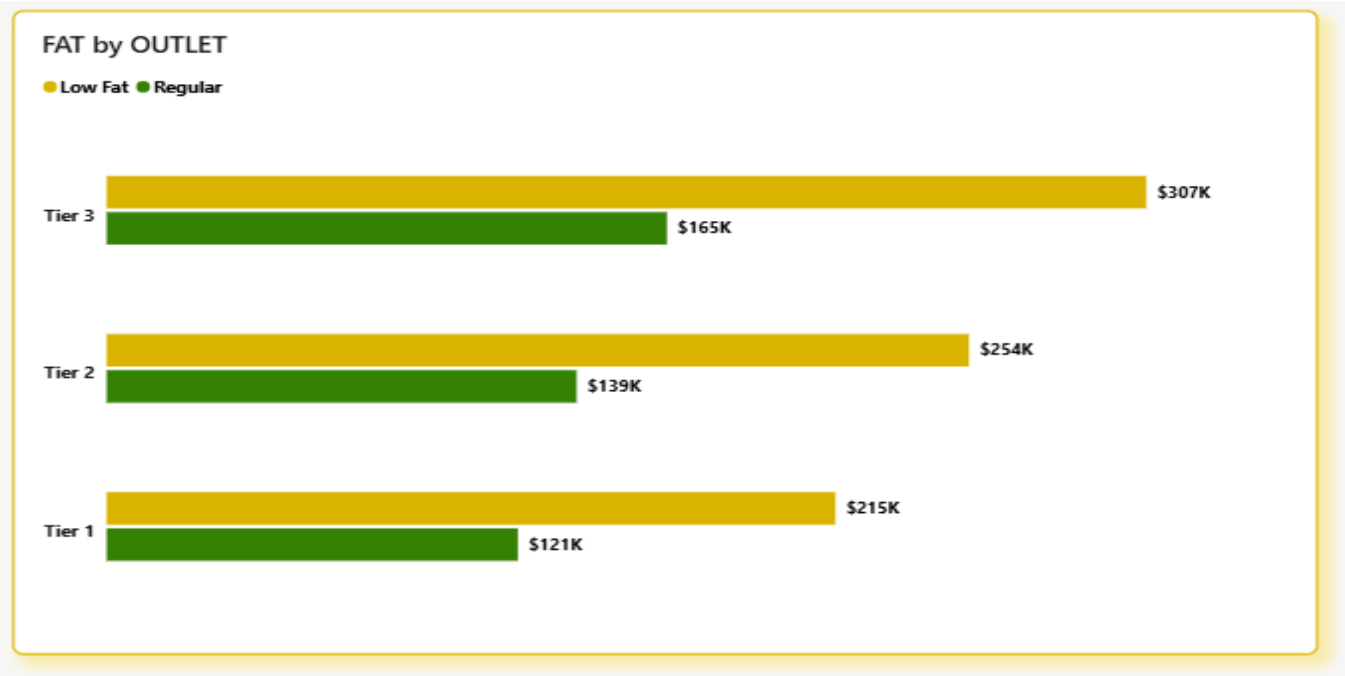

Objective: Analyse the impact of fat content on total sales. Additional KPI metrics: Assess how other KPIs (Avg. sales, No. of items Avg. Rating) vary with fat content. Chart type: Donut Chart.

## 6.2 Total sales by Item Type



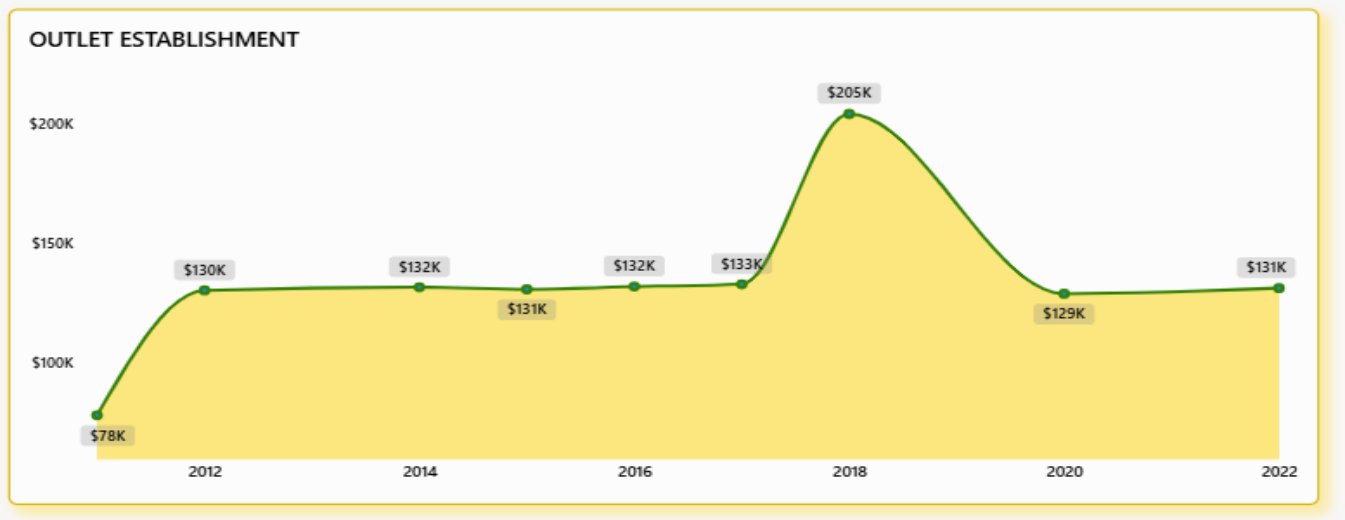

Objective: Identify the performance of different item types in terms of total sales. Additional KPI metrics: Assess how other KPIs (Avg. sales, No. of items Avg. Rating) vary with fat content. Chart type: Bar Chart.

## 6.3 FAT Content by Outlet for total sales



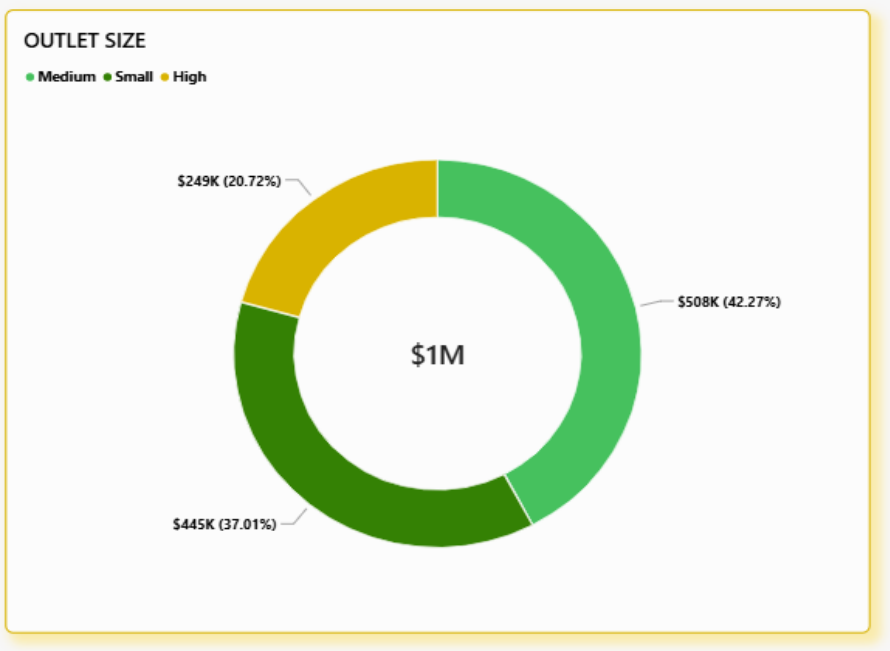

Objective: Compare total sales across different outlet segmented by Fat Content. Additional KPI metrics: Assess how other KPIs (Avg. sales, No. of items Avg. Rating) vary with fat content. Chart type: Stacked Column Chart.

## 6.4 Total sales by Outlet Establishment



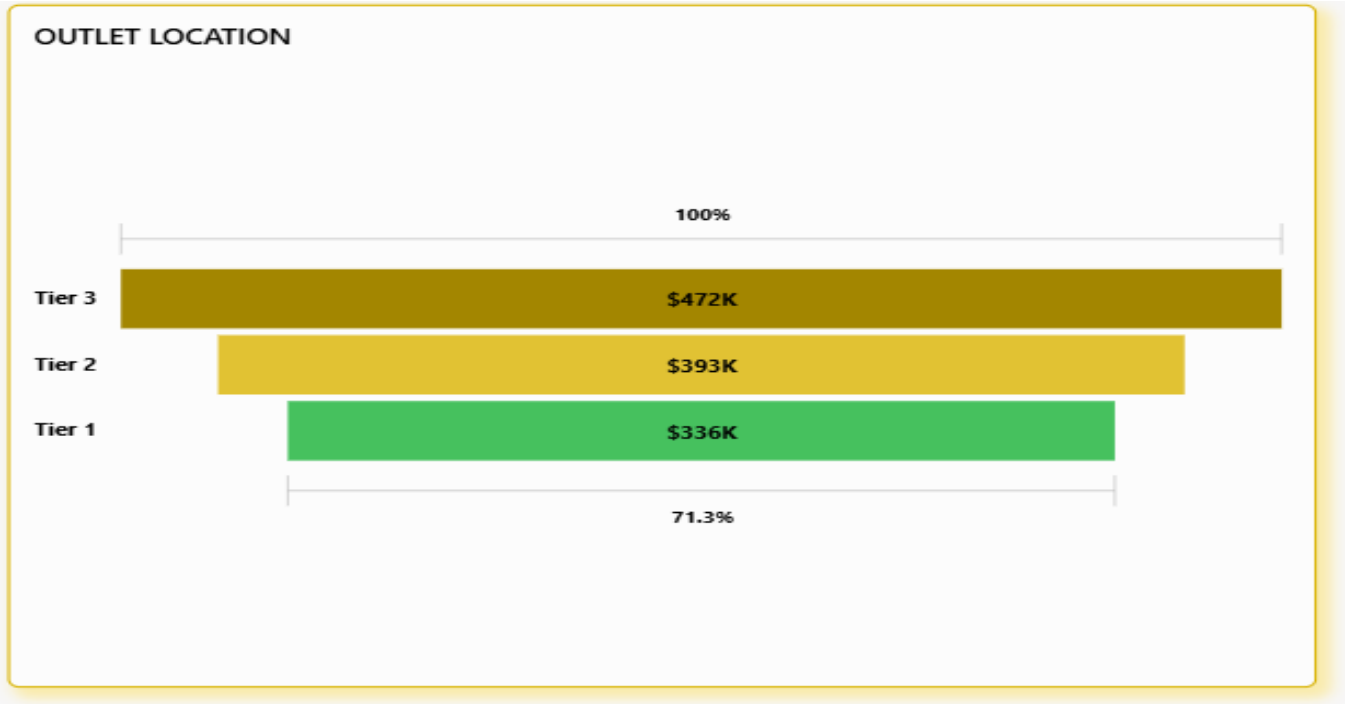

Objective: Evaluate how the Age or Type of outlet establishment influences total sales Chart type: Line Chart.

## 6.5 Sales by Outlet Size



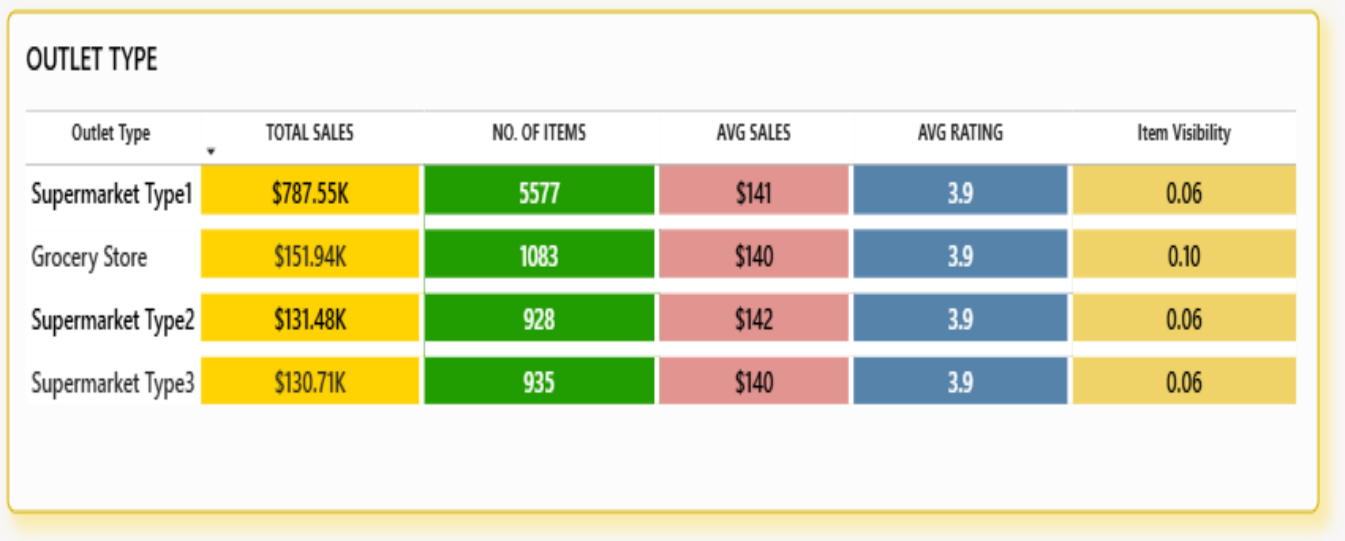

Objective: Analyse the Correlation Between Outlet Size and Total Sales. Chart type: Donut / Pie Chart.

## 6.6 Sales by Outlet Location




**Objective: Assess the geographic distribution of sales across different locations. Chart type: Funnel Chart.**

## 6.7 All Metrics by Outlet Type



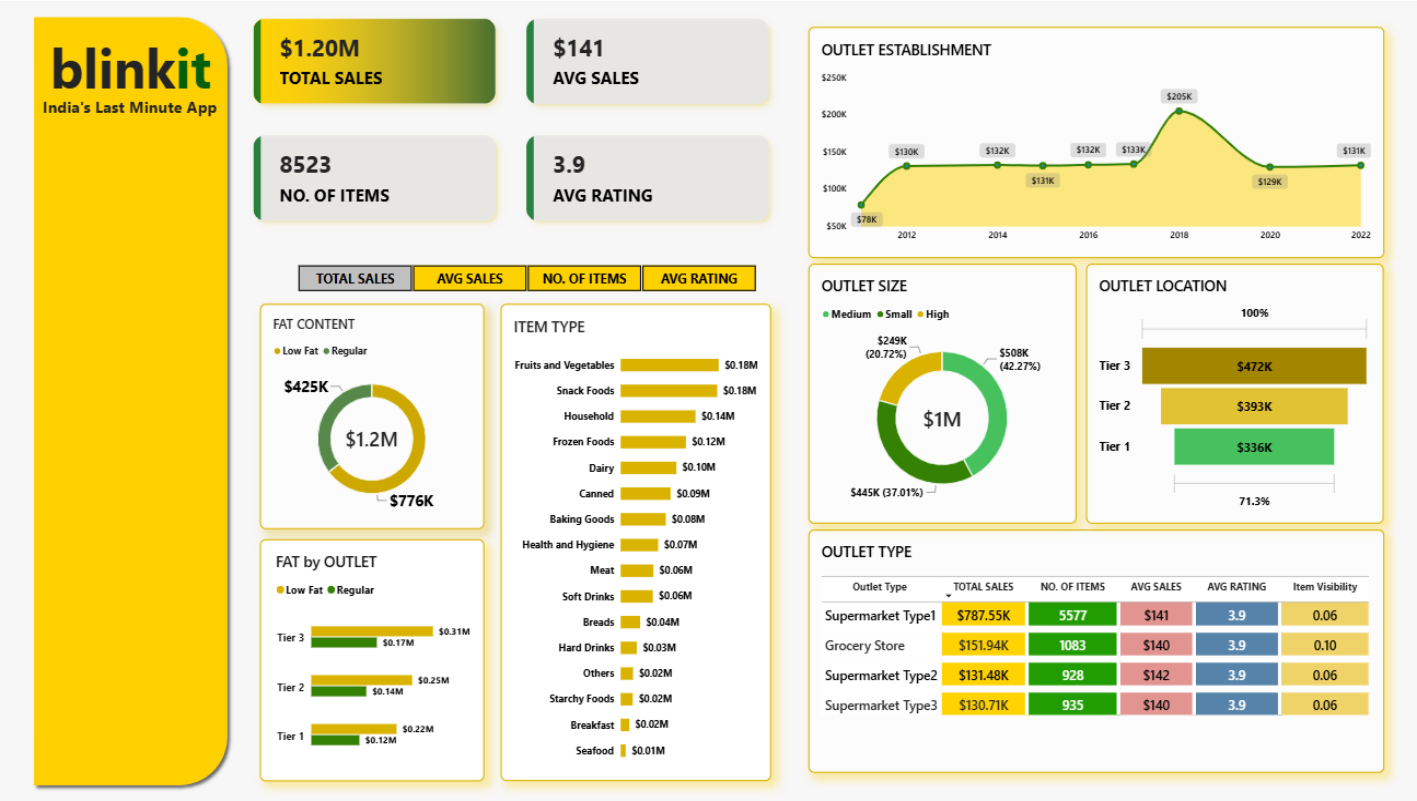

Objective: Provide a comprehensive view of all key metrics (Total sales, Avg. sales, No. of items Avg. Rating) broke down by different outlet types. Chart type: Metrics Card.

# 7. Key Findings

- Low Fat products generated 776,319.69 in sales (64.60%), compared with 425,361.80 (35.40%) for Regular products.
- Fruits and Vegetables generated the highest total sales at 178,124.08, contributing 14.82% of overall sales.
- Supermarket Type1 generated the highest sales for both Low Fat (507,886.30) and Regular products (279,663.59).
- Outlets associated with 2018 recorded the highest total sales: 204,522.26, approximately 17.02% of overall sales.
- Medium-sized outlets generated the highest total sales: 507,895.74 (42.27%).
- Tier 3 locations generated the highest sales at 472,133.03 (39.29%).
- Supermarket Type1 dominates sales, contributing 787,549.89, or 65.54% of total sales.

# 8. Business Recommendations

1. Maintain strong availability of Low-Fat products, given their substantial contribution to total sales.
2. Prioritize shelf availability and replenishment for Fruits and Vegetables and Snack Foods to protect their substantial sales contributions.
3. Ensure adequate inventory of both fat-content categories in Supermarket Type1 outlets, particularly for fast-moving products.
4. Investigate the factors behind the 2018 group's strong total sales, including assortment, outlet format, location, and customer demand.
5. Study the product mix, customer demand, and operational practices of medium outlets to understand their large contribution to total sales.
6. Investigate the demand drivers in Tier 3 locations and assess whether similar product assortment strategies could work in other locations.
7. Protect Supermarket Type1's sales contribution through effective inventory management, replenishment, and monitoring of fast-moving products.




## Appendix: Dashboard Checklist

- Donut chart — Total sales by FAT Content
- Stacked Bar chart — FAT Content by Outlet for total sales.
- Bar chart — Total sales by Item Type
- Line chart — Total sales by Outlet Establishment
- Donut chart — 5 Sales by Outlet Size
- Funnel chart — Sales by Outlet Location.
- KPI cards — Total Sales, Average Sales, Number of Items, Average Rating.
- Slicers — Total Sales, Average Sales, Number of Items, Average Rating.